

Chapter 11: Electricity

11.1 Electric current

Electric current is the rate of flow of electric charge through a conductor. If a net charge Q flows across any cross-section of a conductor in time t , the current is $I = Q / t$. The SI unit of current is the ampere (A). One ampere is one coulomb of charge flowing per second. Current is measured with an ammeter, which is always connected in series with the part of the circuit whose current we want to measure.

11.2 Potential difference

Charges flow only when there is a difference of electric pressure, called potential difference, between two points. The potential difference V between two points is the work done W to move a unit charge Q from one point to the other: $V = W / Q$. Its SI unit is the volt (V). One volt is one joule of work per coulomb of charge. A voltmeter measures potential difference and is always connected in parallel across the points.

Ohm's law

11.3 Ohm's law

The German physicist Georg Simon Ohm found that, at constant temperature, the current flowing through a metallic wire is directly proportional to the potential difference across its ends. This is Ohm's law: $V = I R$, where R is a constant called the resistance of the conductor. The SI unit of resistance is the ohm. A wire has a resistance of one ohm if a current of one ampere flows through it when the potential difference across it is one volt.

Worked example

A bulb draws a current of 0.5 A from a 6 V battery. Its resistance is $R = V / I = 6 / 0.5 = 12$ ohm. If the same bulb is connected to a 3 V battery, the current becomes $I = V / R = 3 / 12 = 0.25$ A.

11.4 Factors affecting resistance

The resistance of a conductor depends on its length, its area of cross-section and the material it is made of. Resistance is directly proportional to length and inversely proportional to the area of cross-section: $R = \rho L / A$, where ρ is the resistivity of the material. Metals and alloys have low resistivity; insulators like glass and rubber have very high resistivity.

Resistors in series and parallel

11.5 Resistors in series

When resistors are joined end to end, they are in series. The same current flows through every resistor. The total potential difference is the sum of the potential differences across each resistor. The equivalent resistance is the sum of the individual resistances: $R_s = R_1 + R_2 + R_3$. Example: resistors of 5 ohm, 10 ohm and 15 ohm in series have an equivalent resistance of 30 ohm.

11.6 Resistors in parallel

When resistors are connected between the same two points, they are in parallel. The potential difference across each resistor is the same, and the total current is the sum of the currents through each branch. The reciprocal of the equivalent resistance is the sum of the reciprocals: $1/R_p = 1/R_1 + 1/R_2 + 1/R_3$. The equivalent resistance is always less than the smallest individual resistance. Household appliances are connected in parallel so each one gets the full supply voltage and can be switched on or off independently.

Heating effect and electric power

11.7 Heating effect of electric current

When a current flows through a resistor, part of the electrical energy is converted into heat. This is Joule's law of heating: $H = I^2 R t$. The heat produced is proportional to the square of the current, the resistance and the time for which the current flows. Electric irons, toasters, heaters and the filament of a bulb use this effect. A fuse is a short wire of low melting point connected in series; it melts and breaks the circuit when too much current flows.

11.8 Electric power

Electric power is the rate at which electrical energy is used. $P = VI = I^2 R = V^2 / R$. The SI unit of power is the watt (W). The commercial unit of electrical energy is the kilowatt hour (kWh), commonly called a unit. $1 \text{ kWh} = 3.6 \times 10^6 \text{ joules}$. Example: a 1000 W heater used for 2 hours consumes 2 kWh, that is 2 units.

Common mistakes

Students often connect the ammeter in parallel; it must be in series. Another frequent error is adding resistances directly for a parallel combination; for parallel resistors you add the reciprocals.